

CHAPTER 3

Separating Mixtures

Lesson 1

Techniques of Separating Mixtures

Do List

1. Demonstrate understanding of the different techniques of separating mixtures.
2. Investigate the techniques of separating mixtures into their individual components.

Energize Your Mind



The Solar salt production is the oldest method of collecting salt. In this method, the sun heats up a pool of sea water. The water evaporates, and the salt is left. as shown in the picture. Impurities in the salt are drained off and discarded. process is inexpensive and is the preferred method of producing salt for people living in coastal areas.

How can you separate the components of a mixture in a simple way? Try the following activity.

Explore Around

SOLAR SEPARATION TECHNIQUES

What You Need

hot water
beaker
salt spoon

What to Do

1. Place 20 ml. of hot water in the beaker. Add salt to the water and stir it using a spoon. Continue adding salt until you see that it is no longer getting dissolved.
2. Leave the salt solution outside where the sun can heat it for at least 5 hours. Note: If the time available to expose the setup to sunlight is limited, your teacher may ask you to leave the setup in the laboratory overnight.
3. After the allotted time, check your setup. Take note of any changes

What You Have Learned

Answer the following questions. Write your answers in the space provided

1. What changes did you observe with your solution after it was exposed to the sun? Describe its appearance.
2. What do you think are the formed particles?
3. What do you think happened to the solution when it was exposed to the sun?

How You Learned

1. What problems did you encounter in this activity?
2. How did you solve the problem/s?

Manual Separation

This technique involves the use of your hands to separate the components of the mixture manually. For example, when you scoop out a spoonful of a mixture of sand and pebbles, you can distinguish the two components of the mixture. By using your hand, you can pick out the pebbles from the mixture until only the sand is left. A mixture of nuts can also be separated using this technique. However, this technique is applicable only for dry mixtures with A mixture of nuts can be separated manually.



the naked eye or for mixtures consisting of liquid and distinguishable solid components. Can you think of other mixtures that can be separated using your hands?

Using a Magnet

In this technique, you can separate components with metallic properties from a mixture by using a magnet. For example, for a mixture of table sugar and iron filings, you can use a magnet to separate the iron filings from the mixture.

Iron is a magnetic substance and is attracted by magnets. When metallic substances such as iron and nickel are mixed with nonmetallic substances such as sugar, a magnet may be used to separate them.

Using a Sleeve

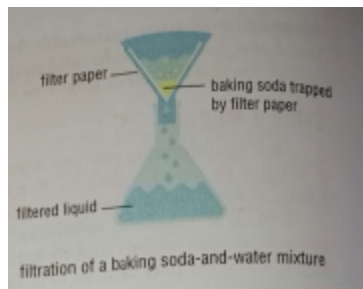
You can also use a sieve to separate the components of a mixture. A sieve is a utensil which is usually made of wire or plastic mesh held in a metal frame. It is often used to separate particles of different sizes. A specialized kind of sieve called a strainer is used to separate solids from liquids. An example of a mixture which can be separated is a mixture of soybeans and flour. You do this by pouring the mixture into a sieve before shaking the sieve sideward to allow the flour to pass through the sieve, leaving only the soybeans behind.

Filtration

This technique uses a filter to separate the components of a mixture such as a mixture of water and baking soda. You can make your own filter by cutting the top part of a plastic bottle or rolling a piece of filter paper into the shape of a cone.

A filter is a device with tiny openings through which a liquid or gas is passed to remove solid particles. Large solids are unable to pass through its holes but liquids and small particles can. This is why filtration is used to separate insoluble solid from a liquid. The insoluble solid called residue, is left in the filter paper, and the liquid, called filtrate, is collected in the container. The

picture shows a liquid being filtered from a mixture of water and baking soda. What happened to the color of the liquid after it was filtered?



Extend Your Understanding

(Word Unlock)

An insoluble substance is one that does not dissolve. For example, sand is insoluble in water.

Decantation

You can also use decantation to separate the components of a mixture. Decantation is a method of removing rust, dirt, sand, and other particles from liquid. The heavier layer with these particles is settled first to separate it from the lighter layer. Decantation can be used for separating mixtures such as gravel and water. Place some gravel into a cup then add water. Once the gravel has settled, pour the water into another cup. Will this method work for a mixture of water and chalk powder or water and oil? Why?



Chromatography

Chromatography is the technique that involves passing a mixture in a solution or suspension through a material that absorbs the components differently. For example, you can separate components of different food dyes by using the molecular interaction of each color with water. You put spots of different color dyes on filter paper and dip the paper in water. Capillary action, the adhesion of water to the filter paper, makes the water travel up the paper slowly and dissolves the different color dyes. The dissolved food dyes travel up the paper and separate into different components. Some of the components stick more to the paper and travel more slowly than the others. Thus, you will see that there are different colors spread out from each of the original food dyes.

chromatography setup

Chromatography has applications in forensic science and pollution monitoring, and is widely used in studying and producing complex mixtures such as food, perfume, and medicines,



Extend Your Understanding (Word Unlock)

Forensic science is the application of science to criminal investigations and involves the collection, preservation, and analysis of pieces of evidence. Forensic scientists may travel to the scene of the crime to collect pieces of evidence such as fingerprints or blood samples, do an autopsy, examine a dead body to determine the cause of death or extent of disease, or perform analyses of the evidence in a laboratory

Using Heat

In Other separation techniques, such as evaporation, crystallization, and distillation, use heating. In evaporation, the solution is heated until all the water evaporates and only the

dissolved substance is left. This technique is used in making salt. crystallization, a solution is heated until it is saturated. A saturated solution contains more of the dissolved material than can be dissolved by the solvent. It is then left to crystallize. This is what happened when you formed salt crystals in the previous activity. In distillation, the mixture is heated to reach the boiling point of one component. Once this happens, that component evaporates and is collected. The distillation of water, for example, involves the heating of water until it evaporates and separates from impurities. The water vapor is then cooled to condense and collected as distilled water.

Some techniques of separating mixtures are manual separation, the use of a magnet or a sieve, filtration, decantation, chromatography, and the use of heat in evaporation, crystallization, and distillation.

Express Your Learning

Answer the following questions:

1. What are the techniques that can be used to separate the components of mixtures?
2. What material can you use to separate sand from water?
3. How will you separate a mixture of oil and vinegar?
4. How did you feel about the lesson? Explain your answer?

Lesson 2

Applications and Benefits of Separating Mixtures

Do List

1. Separate desired materials from common and local products.
2. Give a report on the benefits of separating mixtures from local and common products used in daily activities.

Energize Your Mind



Garbage is a major environmental problem in many places. In Metro Manila alone, 6000 tons of garbage is collected daily. Garbage is transferred to a materials recovery facility (MRF) where it is sorted and segregated into glass, metal, paper, plastic, and organic materials such as leftovers, wood trimmings, fruit and vegetable peels, and manure. Organic materials are

carbon-based matter that are usually living things. Segregation helps to dispose of materials properly and even recycle some of them. Garbage that is not segregated poses problems to people's health and to the environment. from

Very few substances are naturally pure. Most substances are found in mixtures and need to be separated for them to be useful. Try the activity on the next page and see if you can extract the required element.

Explore Around

IRON IN AN UNKNOWN MIXTURE

What You Need

unknown mixture with iron filings magnet

funnel container

filter paper

What to Do

Your teacher will give you an unknown mixture containing iron filings Extract the iron filings using the separation techniques you have learned. Write down each step of your chosen separation method in the box below.

What You Have Learned

Answer the following questions, Write your answers in the space provided.

1. What technique did you use to get the iron filings?
2. Why do you think your technique is appropriate for this mixture? Explain your answer.

How You Learned

1. Is it a mixture? Why use separation techniques in extracting the components of a
2. Will it make a difference if you applied a technique different from what you have learned? Why?

Equip Yourself

Mixtures are found in natural and man-made environments. Do you know how the components of mixtures are separated in nature? How do you shout in other environments, such as in the industrial field?

Natural Separation Methods and Their Benefits

Natural methods also help separate minerals from their sources. For instance, erosion can cause a fragment of gold called a gold nugget to separate from a piece of rock. Flowing water from a stream or river carries the gold nugget to another place such as a brook or a stream. Evaporation separates water from the ground which later on falls back to the ground in its purified form as rain or snow. A wetland is an ecosystem filled with water and aquatic plants. It acts as a natural Alter which removes solid particles from water. Because of the presence of

plants, wetlands slow down water flow. When this happens, suspended particles, metals, and other impurities in the water settle into the sediment layer of wetlands, thus, purifying the water.

Wetland

Soil can also filter water naturally by trapping particles as water goes down through the soil layers. Bacteria and other soil microorganisms further purify water naturally by breaking down its nutrients and contaminants.



Water can be naturally filtered by wetlands and soil.

Domestic and Industrial Separation Methods and Their Benefits

Water needs to be purified not only for human consumption, but for industrial uses as well. Industries need water to produce food, medicines, utilities, and other human needs. Hospitals, schools, and other institutions need purified water to be operational. Through human technology, chemicals, microorganisms, pollutants, contaminants, and hazardous particles can be removed from water using physical processes such as filtration and distillation, natural processes involving soil and wetlands, and the chemical addition of compounds to kill bacteria and other harmful organisms present in water, thus preventing the occurrence and spread of diseases,

Know how to purify water in a simple way by performing the following activity.

Explore Around

SIMPLE WATER PURIFICATION

What You Need

1.5 L soft drink bottle

coffee filter

sand

charcoal

gravel

cheesecloth or any thin cloth

a pair of scissors
water and soil mixture

What to Do

1. Using the materials you have, prepare your own water filter. Illustrate your water filter below.
2. Start purifying the water and soil mixture to test how effective your filter is.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What happened to the unclean water when you poured it over your filter?
2. How did the materials you have used help filter the unclean water?

How You Learned

1. How did you decide on the layers of your water filter?
2. Do you consider your water filter effective? Why?

Aside from water purifiers, coffee makers and air purifiers are household equipment that can be used to separate mixtures. In a coffee maker, the paper filter collects the coffee grounds and allows the brewed solution to pass. On the other hand, air purifiers remove contaminants from the air, such as dust and mites, using filters, activated carbon, or ultraviolet lamps.



Other equipment, such as centrifuge and magnetic separators, are used in research and industries. A centrifuge spins at very high speeds to allow the heavy particles in suspensions to settle very quickly. It is used in laboratories to separate the components of blood. It is also used in food industries to separate cream from milk.



Extend Tour Understanding Know Zonel

A high-speed centrifuge can spin up to 30000 times a minute

Magnetic separators are used in factories. Materials are put on a conveyor belt that is spun by magnetic wheels. As the conveyor belt moves, the nonmagnetic objecta fall off, and the magnetic ones remain



magnetic separator

In the meal industry, iron is extracted from its ore by heating it to a high temperature. The intense heat then burns away substances such as sulfur and oxygen, leaving only the iron. Aside from heating, electricity is used to extract metals such as aluminum. As mentioned earlier, some metals such as gold can be readily found in nature and do not need to be separated from its ore.

However, gold is usually mixed with other substances, such as rock and sand. Gold is separated from



Iron blast furnace

sand by panning because it is denser than sand. In some parts of the Philippines where small-scale mining is being practiced, impure gold is usually mixed with mercury. Mercury can be combined with fine-grained gold to separate gold from sand and rocks. However, mercury is a toxic substance and has negative effects on health. Because of this property, the government has banned the use of mercury. Borax is now considered a safer substitute for mercury.



panning for gold

Extend Your Understanding

(Bridging to Health Mercury is a toxic substance. A person exposed to mercury may experience shortness of breath, vomiting, and vision problems. Long-term exposure to mercury poses problems to the nervous, digestive, and immune systems. It also affects the brain development of fetuses, infants, and children.

Water purifiers, coffee makers, centrifuges, and magnetic separators are examples of equipment that are used to separate mixtures.

Another industry that makes use of separation techniques in oil refineries. Different components of petroleum are extracted by distillation. The components collected on different levels depending on their boiling points. Some of the products of petroleum are gasoline, diesel, and asphalt.

Like oil refineries, perfumeries also use the process of distillation. Sources of oil, such as plants, are soaked in water or alcohol and boiled. The vapor contains oil and water, which are collected and separated using a fractionating column. The cost of a perfume depends on how difficult the essential oil is extracted.

Oil refineries and perfumeries use distillation to create products such as gasoline, diesel, and perfume.

Express Your Learning

Answer the following questions:

1. What are some household devices that are used to separate mixtures?

2. How does distillation help in separating the components of petroleum?

How does the community benefit from the use of separation techniques in industries?

How did you feel about the lesson? Explain your answer.

Map It

MIXTURES

can be separated by

manual means

filtration

chromatography

use of

decantation

heating

sieve

magnet

evaporation

can be through

crystallization

Distillation

Wrap Up!

Mixtures can be separated into their different components by using your hand, a magnet, or sieve.

Other separation techniques include filtration, decantation, chromatography, and the use of heat in evaporation, crystallization, and distillation.

Water is naturally filtered by wetlands and soil.

A water purifier, coffee maker, centrifuge, and magnetic separator are some equipment that are used to separate mixtures.

Oil refineries and perfumeries use distillation to create products such as gasoline, diesel, and perfume.

Size Up!

Knowledge

Encircle the letter of the correct answer.

1. Mixtures can be separated into their components by
a. observation techniques

b. chemical process

c. scientific investigations

d. physiological assessment

2. Charcoal powder is insoluble in water. When water is filtered, the powder becomes the الرس
a. filtrate

b. residue

c. solute

d. solvent

3. A separation technique that
a. filtration

b decantation

Chromatography

d distillation

4. A magnet may be used to separate soil from

a. wood shearings

iron nails

plastic buttons

4 salt

5. Petroleum can be separated into different types of oil by distillation bonus

a. it has a low boiling point

h. it has a high boiling point

c. different liquid components boil off at different temperatures

d. all its liquid components have the same boiling point

Process/Skill

Encircle the letter of the correct answer.

1. Which can help you determine the appropriate separation

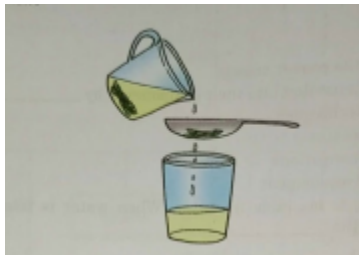
a. proportion in which the components are joined together

B. color of the mixture

C. solubility of the components

d. time available for separation:

For numbers 3-4 A mixture of fresh milk and substance X is separated by the method shown below.



2. What technique is shown?

a. filtration

b. Decantation

3. What could be substance X?

c. chromatography

d. distillation

tea leaves

d. Alcohol

4. John wanted to separate the sugar component of fresh milk. He planned to apply the crystallization technique to do this. Do you agree with John's plan?
chemically combined.

- a. No Sugar cannot be separated because the components of milk are
- b. No. Evaporation is a better technique,
- c. Yes. Crystallization will burn the sugar.
- d. Yes. Crystallization is appropriate for soluble substances such as sugar.

& An experiment includes the following materials: Bunsen burner, test tube, and test tube holder

5. What could be the separation technique used in the experiment?

- a. filtration
- b. decantation
- c. evaporation
- d. chromatography

6. Which of the following pairs can be used to separate the components of ink?

- a. a wire mesh and water
- b. evaporating dish and Bunsen burner
- c. beaker and stirring rod
- d. coffee filter and alcohol

7. Desalination is the process of obtaining drinking water from seawater. Which of these techniques could not be used in a desalination plant?

- a. distillation
- b. evaporation
- c. crystallization
- d. condensation

8. Which of the following steps should not be followed in doing chromatography?

- a. Put a small amount of ink on the chromatography paper.
- b. Completely submerge the chromatography paper in water.
- c. Observe if the ink separates into different colors
- d. Dry the chromatography paper after the ink has separated completely or has reached the top.

9. In which mixture is it appropriate to use a magnet to separate its components
iron and sand

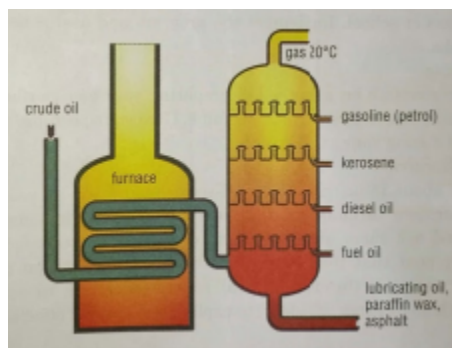
- b. iron and sugar
- c. copper and wood
- d. copper and gold

10. Shown in the picture are the products of petroleum from distillation. Which statement is true
based on the image?

crude oil

ful of Intricating oi

- a. Gas should be more expensive than diesel oil.
 - h. Asphalt will most likely be collected in gaseous form.
- Fuel oil is more difficult to collect and purify than gasoline
- d. Lubricating oil is not an important product



Understanding

Answer the following questions:

1. How can you determine the best way to texture?
2. Alcohol was accidentally mixed with water. Propose a process to separate the two components.
3. What are the benefits of learning about the techniques in separating mixtures?

Product

Do the following tasks:

1. Garbage Segregation Program

As a group, propose a way by which you can properly segregate your garbage in class or school. Implement the program and assess its effectiveness after two weeks.

2. Perfume Making

As a pair, research on a method of preparing your own perfume. Upon the approval of your teacher, make your perfume. Give your perfume a brand name and present it during your exhibit day.

3. Community Research

Find out about industries or small-scale businesses in your community that use separation techniques. Interview key persons, including owners or employees, and ask them about the kind of separation techniques that they use, the equipment that they have, and the merchandise that they produce. Present your research through an oral report. Feel free to use any visual materials, such as pictures or videos, to explain your report effectively.

Living Things and Their Environment

UNIT

2

Have you ever wondered how living things survive and interact with the environment? How do plants breathe? How do animals obtain their food? How does the environment sustain all living organisms

Look at the pictures above. You can see people, plants, and animals. Each of these living things is remarkable in its own way. For example, plants can produce their own food. Animals can move on land, in the water, and in the air. They may have simple systems such as the jellyfish or worm. Human beings and animals have body systems that play specific roles in keeping that alive. All these living things interact with each

Luther and the environment. In this unit, you will learn about living things as well as the environment where they live. You will also learn about the different body systems of human beings.



The organizational chart below shows how all aspects of learning in this unit are connected:

Big Ideas

The Human Body

Animals

Plants

Ecosystems

Performance Standards

Make a chart that shows

healthful habits that promote the proper functioning of human body systems.

Make an inventory of vertebrates and invertebrates that are commonly seen in the community.
are

Practice ways of caring for and protecting animals.

Make a multimedia presentation on how the parts of the reproductive system of spore-bearing and cone-bearing plants ensure their survival.

Create a flyer about plant props

Conduct a group discussion on issues involving the protection and conservation of ecosystems,

Essential Understanding

Now organs work together to form an organ system.

Understand the characteristics and classifications of plants and animals.

Understanding the interactions among the components of different ecosystems.

Essential Questions

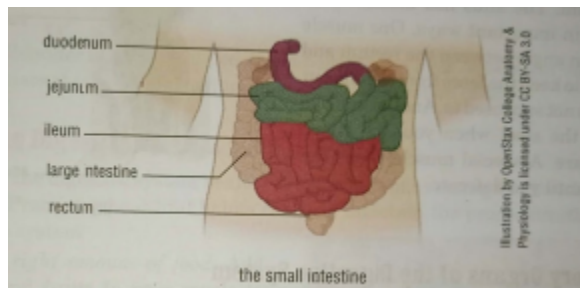
Why do you need to maintain proper body functioning?

Why is it important to conserve and protect the plants and animals?

Why is it important to know about the interactions among the living things in ecosystems?

Small Intestine

The small intestine is a long tubular organ that connects the stomach and the large intestine. It is about 6 meters long and loosely coiled in the abdomen. The small intestine is the longest part of the alimentary canal. It is called "small" because it has a small diameter of about 2.54 cm or 1 inch. It is the region where most of the digestion and absorption occurs.



The small intestine has three regions, the duodenum, jejunum, and ileum. The thick fluid from the stomach moves to the duodenum, the first part of the small intestine. It works with pancreatic juices and bile from the gallbladder and liver, and breaks down the food into a form where nutrients can be extracted. The jejunum and ileum are responsible for the absorption of nutrients from the food into the bloodstream. J

Extend Your Understanding (Word Unlock)

Bile is a bitter-tasting fluid that is produced in the liver. It helps in the digestion of fats and proteins.

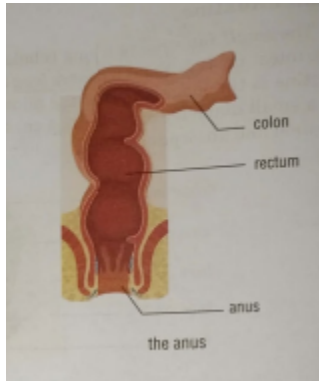
Large Intestine

Aten

From the small intestine, where most of the food and nutrients have been absorbed, waste materials are moved to the large intestine or colon by peristalsis. As the waste product passes through the large intestine, water is absorbed, and stool, which consists of undigested food and waste products, is formed. Another name for stool is feces. The feces is then deposited in the rectum and eliminated from the body through the anus in a process called defecation.

Anus

The anus is the last part of the alimentary canal. It is the opening through which solid wastes are expelled. The anus is located at the bottom of the rectum. The anus has muscles that function in important ways. One muscle creates an angle between the rectum and the anus to keep the feces from coming out when it is not supposed to. Another muscle controls the stool when you are asleep or unaware. A special muscle keeps the stool in until you defecate.



Accessory Organs of the Digestive System

An accessory digestive organ is an organ that aids in food digestion but is not part of the alimentary canal. These organs are the pancreas, liver, and gallbladder. These organs work together to digest, process, and absorb food.

Liver

The liver is a large organ on the upper right side of the abdomen. It is protected by the rib cage. The liver secretes bile, which helps the small intestine break down and absorb fats, cholesterol, and nutrients, making them easy to digest. It also stores and processes carbohydrates, which are released when energy is needed. The liver also filters the blood from the digestive tract before it is circulated in the body. Because the liver has many important functions in the body, it has the ability to undergo regeneration, or the process of renewal, restoration, and growth of new tissue.

pancreas

the liver



Pancreas

nu

The pancreas is a gland in the abdomen that is about 15 cm long. It is surrounded by the stomach, small intestine, spleen, and gallbladder. The cells of the pancreas secrete and release digestive juices into the small intestine to help it in the chemical digestion of food into nutrients.

The pancreas also helps in regulating blood sugar.

Gallbladder

The gallbladder is a pear-shaped organ beneath the liver. It receives and stores bile and releases it into the duodenum to help in the digestion of fats.

Care for the Digestive System

Keeping the digestive system healthy is as important as caring for the other parts

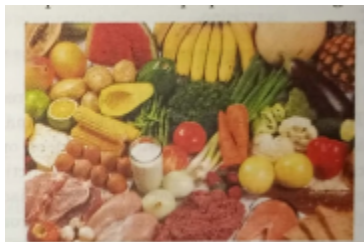
Care for the Digestive System

Keeping the digestive system healthy is as important as caring for the other parts of the body.

Practicing healthful habits can help maintain the proper functioning of the digestive system.

Eat the right amount of food. Add vegetables and fruits to your meals to improve your health.

Eat your food well. How you eat is just as important as what you eat. Thus, eat slowly, chew your food well, and avoid overeating.



healthy foods



drinking enough water



exercise

Always stay hydrated. In all cases, your body needs enough fluid or water to function well. Drink water whenever you feel thirsty.
drinking enough water

Exercise for at least 30 minutes a day.

This does not necessarily mean that you need to go to a gym. The simple activities that you do, such as walking, running, and doing household chores, can already be considered as forms of exercise.

The digestive system is a system of organs responsible for converting food into nutrients that are absorbed and used by the body to keep all its systems working. It is also responsible for the removal of food that cannot be converted, absorbed, or used by the body.

Food is digested by a combination of two methods: mechanical digestion and chemical digestion.

The digestive tract or alimentary canal is the continuous muscular tube that carries food through different parts of the digestive system

The digestive tract is composed of the mouth, pharynx, esophagus, stomach, small intestine, large intestine, and anus

Caring for the digestive system involves healthy habits such as eating the right amount of food, eating your food well, staying hydrated, and exercising.

The digestive system supplies the rest of the body with the nutrients it needs. Therefore, when the digestive system is unable to function well, the other organs are affected. For example, the lack of food or nutrition may make a person feel weak, causing the muscular system to not function well. The digestive system also contains one hundred million nerve cells. Any discomfort or pain in the system is relayed to the brain. The pain can keep a person from concentrating or doing simple tasks.

Perform the activity below and learn more about the digestive system.

Explore around

TRACING THE PATH OF FOOD

What You Need

diagram of the digestive system be provided by the teacher

What to Do

1. Trace the imaginary path of food as it passes through the different parts of the digestive system. Mark this path on the diagram using a pencil.

2. Label the following parts:

- a. mouth
- b. pharynx
- c. esophagus
- d. stomach
- e. small intestine

f. large intestine

anus

h. liver

1. pancreas

J. gallbladder

3. Draw a circle around the part where mechanical digestion of food takes place.

4. Draw an arrow toward the part where the absorption of nutrients occurs.

5. Draw a square on the part where the absorption of water occurs.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What structures are involved in mechanical digestion?

2. What structures are involved in the absorption of nutrients and water?

3. Why is absorption of water important?

How You Learned

1. Do you think another pathway for food would keep the body functioning properly? Why or why not?

2. Did the lesson help you look forward to maintaining a healthy digestive system? How?

Express Your Learning

Answer the following questions:

1. What organs are involved in digestion? What are their functions?

2. How do the organs you described in item 1 work together?

3. How is the small intestine different from the large intestine?

4. How does the digestive system interact with your other body systems?

5. How did you feel about the lesson? Explain your answer.

Lesson 4

Respiratory System

Do List

1. Name the parts of the respiratory system.

2. Identify the functions of the respiratory system.

3. Explain how the organs of the respiratory system work together with other organ systems.

4. Enumerate ways of taking care of the respiratory system.

Energize Your Mind

Look at the picture above. What do you think the child will feel after jogging? He may probably feel energized after jogging and taking in deep breaths of air. Have you ever wondered how the air from the environment enters and leaves the body?

Equip Yourself

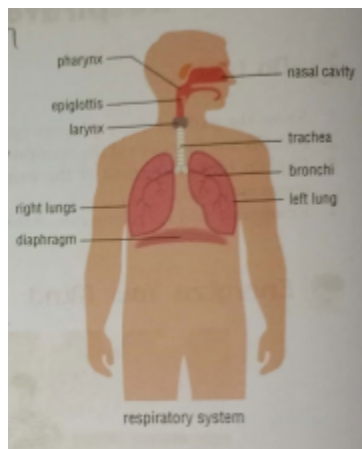
After a vigorous activity such as running or jogging, it is normal for the body to experience an increase in breathing rate. This condition results from the work exerted by the lungs to allow the body to receive oxygen and circulate oxygen-rich blood through it. After a few minutes, the breathing evens out, and the body returns to normal functioning.

Breathing is one of the vital body processes. You do not notice it, but while resting you inhale and exhale 12 to 20 times a minute, day after day.

This is one of the functions

The respiratory system is a system of specific organs and structures that provide oxygen to the cells of the body and remove carbon dioxide from the cells. This involves two processes: breathing and respiration. Breathing is the physical process of inhaling and exhaling air into and out of the lungs. Respiration is a chemical process that involves the movement of oxygen from the environment to the cells within the body as well as the transport of carbon dioxide from the body to the environment.

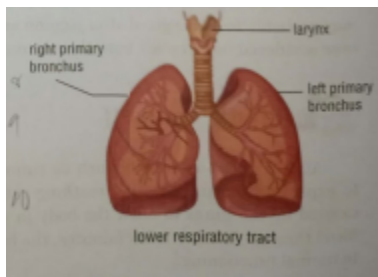
The respiratory system is divided into two main parts: the upper respiratory tract and the lower respiratory tract. The upper tract consists of the nose and nasal cavities, the pharynx, and the larynx. The lower tract includes the trachea, bronchi, bronchioles, and lungs.



Parts of the Respiratory System and their Functions

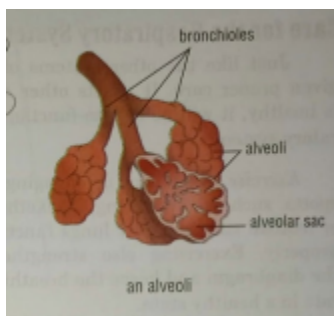
The nose is the part of the respiratory system that filters and moistens air. The pharynx serves as the passageway of air as it moves incoming toward the lungs. The larynx or voice box is the part that is responsible for breathing and sound production. It contains the epiglottis, which is the part that prevents food from entering the tract that goes into the lungs.

The trachea, also known as the windpipe, carries the air from the larynx down to the bronchi. The bronchi are two air passageways that lead to the bronchioles. The bronchioles are the small branches of the bronchi thus also serve as passageways of air.



Attached to the bronchi and bronchioles are the lungs, the main organ of the respiratory system. The alveoli or air sacs are found in the lungs. The alveoli are compartments where exchange between oxygen and carbon dioxide occurs.

The diaphragm, which is located underneath the lungs, is the main respiratory muscle. During inhalation, the muscles in the diaphragm contract and pull toward the abdominal cavity. This process enlarges the thorax and allows air to inflate the lungs.

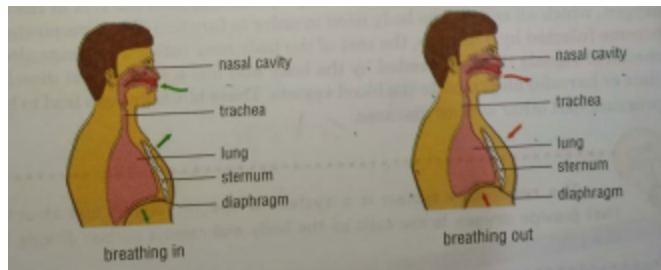


Extend Your Understanding (Know Zone)

Hiccups are involuntary contractions of the diaphragm. Each contraction is followed by a sudden closure of your vocal cords, which produces the characteristic "hic" sound.

Process of Breathing.

Breathing involves two main processes, inhalation and exhalation. Inhalation is the process of taking in air into the lungs. During this process, the diaphragm contracts, causing the chest to expand and the inner air pressure to decrease. Air rushes into the lungs. Thus, Meanwhile, exhalation is the process by which air is released from the lungs. During this process, the diaphragm relaxes, causing the chest to decrease in size. This condition allows the air to flow out as the lungs deflate.



Care for the Respiratory System

The system should be just like the other systems of the body, the respiratory system given proper care. It affects other organs in the body, the respiratory system should be healthy. Jay System? Everyta. Exercising or engaging in sports such as swimming, basketball, or football can help your lungs function properly. Exercising also strengthens the diaphragm and keeps the breathing rate in a healthy state.



balanced diet

Maintain a healthy diet. A healthy diet is important because it is the source of the nutrition and energy needed by the body. You need to consume a balanced diet that includes sufficient amounts of proteins, carbohydrates, fats, vitamins, and minerals.

Avoid smoking. Smoking involves the use of harmful substances that can affect many organs, especially the lungs. You should also stay away from smoky places to avoid being exposed to secondhand smoke, which can cause respiratory diseases.

Go for regular medical checkups. Get medical checks yearly to monitor the condition of the respiratory system as well as the other body systems. The doctor will advise you on how you can keep your respiratory system healthy

As already mentioned, the respiratory system supplies the rest of the body with oxygen, which all cells in the body need in order to function. When respiratory organs become infected by a

disease, the rest of the body may suffer. The lungs also activate chemical signals that are needed by the body and have a system that dissolves blood clots or harmful blockages to the blood vessels. These blockage can lead to high blood pressure and other serious diseases. The respiratory system is a system of specific organs and structures that provide oxygen to the cells of the body and remove carbon dioxide from the cells.

3. Get the data of two of your classmates. Compare and analyze the results.

Name _____

Breaths/Minute While Sitting, _____

Breaths/Minutes After Jogging _____

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. How does exercise affect the rate of breathing?
2. How do you compare your breathing rates while sitting and after exercising?

How You Learned

1. Does the activity make sense to you? How?
2. What is the importance of this activity to your respiratory system?

Express Your Learning

Answer the following questions

1. How would you explain the inhalation and exhalation processes?
2. How can you maintain a healthy respiratory system?
3. How did you feel about the lesson? Explain your answer

Lesson 5

Circulatory System

Do List

1. Name the main parts of the circulatory system
2. Identify the functions of the circulatory system
3. Enumerate ways of making care of the circulatory system
4. Explain how the major organs of the circulatory system work together with other organ systems

Energize Your Mind



Pause for a moment and hold your wrist with your middle and index fingers. What do you feel? What you feel is your pulse. Pulse is the rate at which the heart beats. You can feel your pulse when you touch certain parts of your body, such as your wrist, neck, and groin. How does the heart beat? How does the blood circulate? How does the blood travel through your body? These processes involve the work of the circulatory system.

Equip Yourself

The circulatory system, also called the cardiovascular system, is a group of organs or structures that is responsible for circulating blood and transporting nutrients and oxygen throughout the body. It also removes substances and waste products that are not needed by the body.

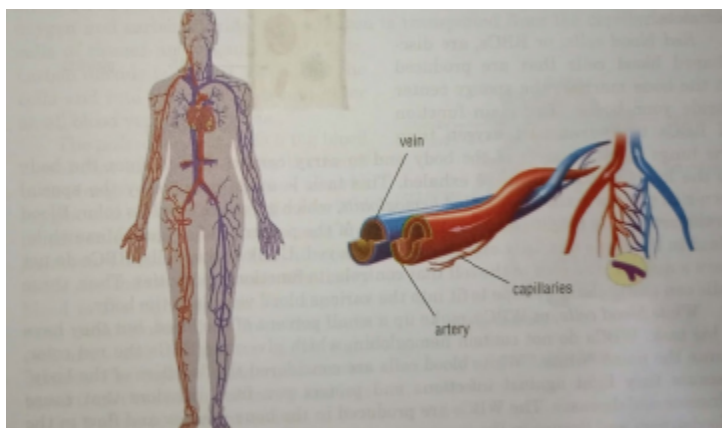
Parts of the Circulatory System

The human circulatory system is composed of three major parts: blood, blood vessels, and heart.

Blood circulates all throughout the body through a series of hollow tubes called blood vessels. These are the arteries, capillaries, and veins

Arteries have thick walls that stretch and carry blood away from the heart. The walls of the artery can stretch and are usually strong enough to withstand the pressure of the blood flowing. Veins are blood vessels that carry blood back to the heart. As blood travels through the veins, the valves in the veins keep the blood from flowing backward.

Capillaries are tiny blood vessels that allow the exchange of gases between the body cells and blood. The capillaries take nutrients, oxygen, and other materials to all cells in the body and to remove their waste materials.

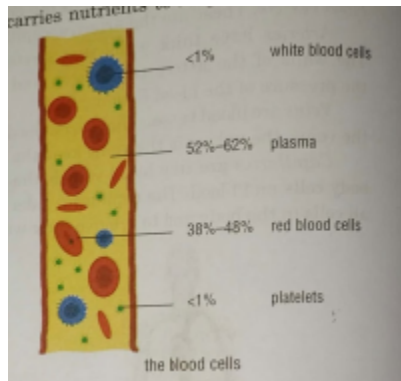


Extend Your Understanding (Word Unlock)

A valve is a small piece of tissue in the heart or in veins that controls the flow of blood and helps it flow in only one direction.

Blood is both fluid that brings nutrients, oxygen, and other substances to the cells and transports waste materials away from those cells. A human body on average has about 4.5 to 5.5 liters of blood, an amount that can fill nearly half a bucket. Blood is called the river of life" that flows through the body because all parts of the body, and second, it fights against infections.

white blood c



Blood is composed of plasma and formed elements. Plasma is the liquid portion of the blood. It is clear and yellowish. It transports blood cells, along with nutrients, hormones, antibodies, and proteins, throughout the body. Plasma also helps remove wastes from the body.

The formed elements in the blood are red blood cells, white blood cells, and platelets

30%-48%

red blood cells

41%

platelets

the blood cells

Red blood cells, or RBCs, are disc-shaped blood cells that are produced in the bone marrow, the spongy center inside your bones. The main function of RBCs is to transport oxygen from the lungs to other parts of the body and to carry carbon dioxide from the body to the lungs so that it can be exhaled. This task is made possible by the special iron-rich protein in the blood called hemoglobin, which gives the blood its color. Blood in the arteries is bright red in color because of the presence of oxygen. Meanwhile, blood in the veins has little oxygen and is dark red. Unlike other cells, RBCs do not have a nucleus, the part of the cell that controls its functional activities. Thus, these

cells can easily change shape to fit into the various blood vessels in the body. White blood cells, the WBCs, make up a small percent of the blood, but they have a long task. WBCs do not contain hemoglobin, which gives the RBCs the red color, hence the name "white." White blood cells are considered the "soldiers of the body" because they fight against infections and protect you from invaders that cause illnesses and diseases. WBCs are produced in the bone marrow and flow in the bloodstream and tissues as part of the immune system to fight microorganisms that

invade your body and threaten your health. When the body is under attack, WBCs rush in to destroy the harmful substances and prevent illness.

Platelets are called fragments that are shaped like small plates. These formed elements are involved in blood clotting. They are the smallest of the blood cells and can only be seen under a microscope. When you are wounded, you bleed because blood vessels are opened. As soon as the bleeding starts, platelets travel to the damaged area and change form by growing sticky tentacles that help them stick to one another and clump together in the damaged area to control the bleeding. When needed, platelets send out signals to attract more platelets to pile onto a clot.

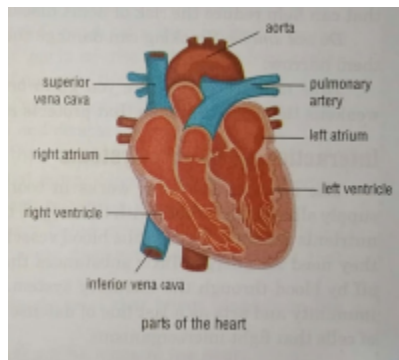
The heart is a muscular organ that pumps blood throughout the body. It is about the size of a fist. It is located in the chest, between the lungs, behind the breastbone, and above the diaphragm. The walls of the heart are composed of cardiac muscle.

As the heart contracts, the cardiac muscle becomes smaller, and the blood flows into the arteries. When the heart relaxes, it fills up with blood from the veins. One complete heart contraction and relaxation is called a heartbeat.

The heart has two upper chambers and two lower chambers. The upper chambers are the right atrium and the left atrium. These chambers receive the blood that enters the heart. The bottom chambers are the right ventricle and the left ventricle, which pump blood out of the heart.

Blood that is carried to the heart absorbs oxygen from the lungs and transports it to the body.

The aorta is the main artery of the body that supplies oxygenated blood to the circulatory system. The aorta branches into arteries, which branch small blood vessels that extend from arteries and lead to capillaries through which oxygen and carbon dioxide pass. Carbon dioxide is transported from the capillaries to the cells.



cells of tissues and organs. Meanwhile, carbon dioxide is carried away from the cells and into the capillaries and enters small blood vessels and veins.

The pulmonary artery is a big blood vessel that brings blood from the heart to the lungs. At the lungs, blood absorbs oxygen and releases carbon dioxide. The blood returns to the heart through the pulmonary veins. Meanwhile, the vena cavae are the two large veins that carry blood to the right atrium. The superior vena cava carries blood from the brain and arms and into the top of the right atrium. The inferior vena cava carries blood from the abdominal cavity and legs to the bottom of the right atrium.

Care for the Circulatory System

How can you care for the circulatory system? Follow those tips to have a healthy and well-functioning circulatory system. Exercise regularly. Exercise strengthens your heart and lowers your blood pressure. Avoid eating too much fried food. Those contain unhealthy fats or fats. Digesting these substances will produce unwanted by-products that may eventually be deposited in the blood vessel. On the other hand, boiling or steaming food reduces the amount of oil or fats. There are also oils that are more beneficial to health than others. Oil from corn, soya, or sunflower contains substances that are good for the heart. By contrast, margarine and other butter substitutes contain unhealthy fats. Eating these types of food could lead to heart diseases and therefore should be avoided. Include fish in your diet. These foods are less fatty compared to meat. Also, eat more vegetables and fresh fruits because they contain fiber, vitamins, and minerals that can help reduce the risk of heart disease. Do not smoke. Smoking can damage the walls of the blood vessels and can make them narrow. Take time to relax before you go to bed. This helps refresh your mind. Stress weakens the defense system that protects against diseases.

Interaction with Other Systems

The circulatory system works in coordination with the respiratory system to supply all cells in the body with the oxygen that they need. From the digestive system, nutrients also flow through the blood vessels to give all the organs the materials that they need for energy. Toxic substances that have built up in the body are filtered off by blood through the excretory system. The circulatory system also helps build immunity and acts as a last line of defense against infection because of the presence of cells that fight microorganisms.

partial partial x (x) = partial partial x (x) partial partial x (x)

The circulatory system is a group of organs or structures that is responsible for circulating blood and transporting nutrients and oxygen throughout the body. It also removes substances and waste products that are not needed by the body.

The human circulatory system is composed of three major parts: blood vessels, blood, and heart.

Blood vessels are a series of hollow tubes through which blood passes.

Arteries are blood vessels with thick walls that carry blood away from the heart.

Veins are blood vessels that carry blood back to the heart.

Capillaries are tiny blood vessels that allow the exchange of gases between the body cells and blood.

Blood is a body fluid that brings nutrients, oxygen, and other substances to the cells and transports waste materials away from those cells.

Blood is composed of plasma, red blood cells, white blood cells, and platelets.

Plasma is the liquid portion of the blood.

Red blood cells are disc-shaped blood cells that transport oxygen. From the lungs to the other parts of the body and carry carbon dioxide from the body to the lungs so that it can be exhaled.

White blood cells fight against infections and protect you from invaders that cause illnesses and diseases.

Platelets are cell fragments that are involved in blood clotting.

The heart is a muscular organ that pumps blood throughout the body.

One complete heart contraction and relaxation is a heartbeat.

The heart has two upper chambers called atria that receive the blood that enters the heart and two lower chambers called ventricles that pump blood out of the heart.

The pulmonary artery is a big blood vessel that brings blood from the heart to the lungs

The pulmonary veins bring blood from the lungs to the heart. The vena cava are the two large veins that carry blood to the right atrium.

The superior vena cava carries blood from the brain and arms to the top of the right atrium. The inferior vena cava carries blood from the abdominal cavity and legs to the bottom of the right atrium.

Eating appropriately and leading a healthy lifestyle make the circulatory system function properly

Explore Around

THE "TIRELESS" HEART

What You Need

pin

rubber ball

Water

Basin

What to Do

1. With the pin, make a small hole at the top of the rubber ball.
2. Fill the basin with water. Put the ball under the water and squeeze it until it is filled with water.
3. Take the ball out of the water. Squeeze it several times for a few seconds. Observe what happens.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What happened each time you squeezed the ball?
2. Did the shape of the ball change every time the water was squeezed out? Why?
3. What is the connection between the activity and the lesson that you have learned about the circulatory system?

How You Learned

1. How did the activity make sense to you?
2. Did the activity clearly show how the heart works? Why do you say so?

Express Your Learning

Answer the following questions:

1. What are the three main components of the circulatory system?
2. What is the composition of blood?
3. How would you explain blood circulation?

4. How do the circulatory organs work together as a system?
5. How did you feel about the lesson? Explain your answer.

Lesson 6

Nervous System

Do List

1. Name the parts of the nervous system.
2. Identify the functions of the nervous system.
3. Enumerate ways of taking care of the nervous system.
4. Explain how the major organs of the nervous system work together with other organ systems.

Energize Your Mind

From the image above, can you tell the kind of activity that is being done? You should be able to see that there is more to just playing basketball. Every part of your system plays the game. You dribble, guard, run, and do other actions, such as making strategies for your team to score points and win the game and displaying values such as sportsmanship and fair play. These are reactions and responses that you are expected to put in the game. What are the body systems involved whenever you play basketball or other sports activities? All such activities are coordinated and controlled by the nervous system.

